DATA ENGINEERING

Data Warehouses & Lakehouses

Where analytics lives:
from star schemas to open table formats.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

OLTP vs OLAP

OLTP
transactions

- Many small writes
- Row-oriented, normalized
- Current state
- Ex: Postgres, MySQL

OLAP
analytics

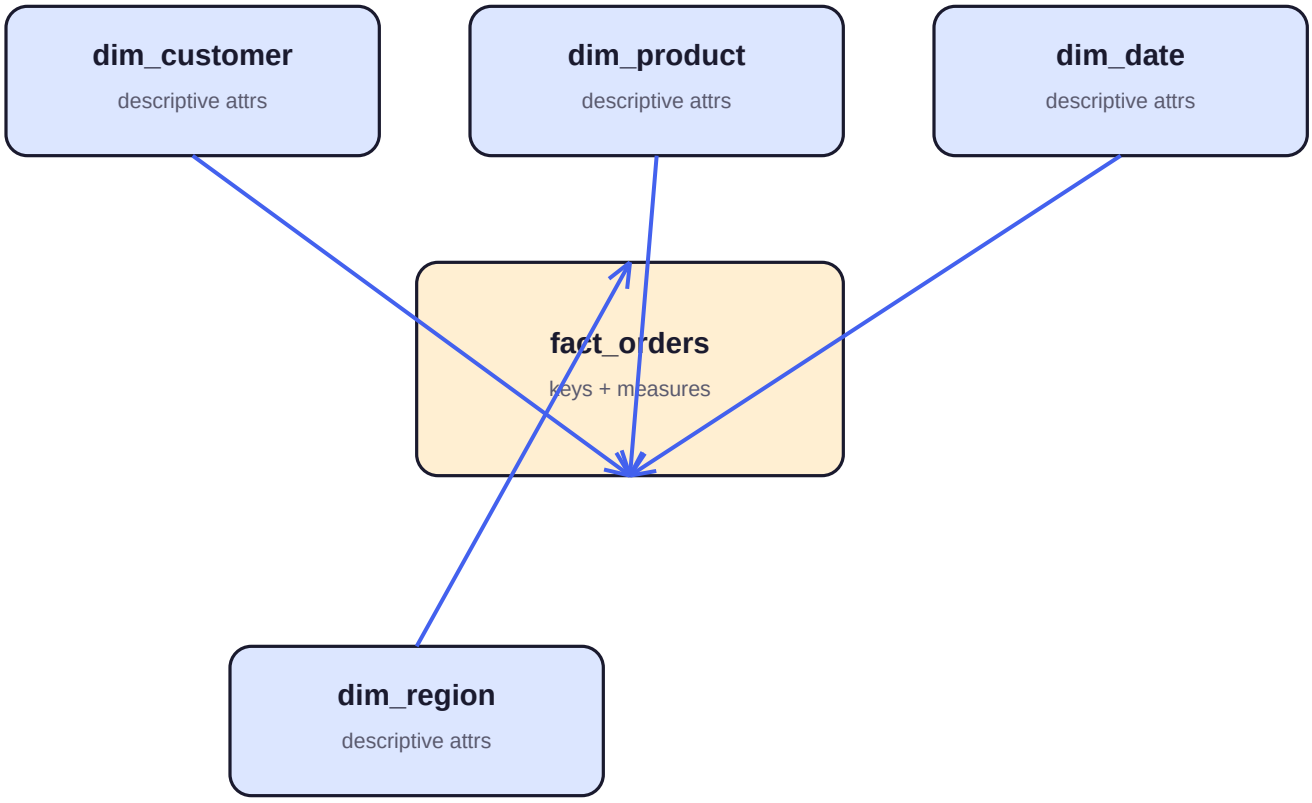
- Few huge reads
- Columnar, denormalized
- Historical trends
- Ex: Snowflake, BigQuery

Never run analytics on your production OLTP — one bad query locks the checkout flow. Replicate to OLAP.

STEP 2/6

Star Schema: Facts & Dimensions

The lingua franca of analytics modeling.



Facts: measurable events (revenue, quantity). Dimensions: who/what/when/where.

Denormalized dimensions = fast joins = happy analysts.

STEP 3/6

Columnar Storage: Read Less, Go Faster

Analytics touches few columns of many rows — store by column.

Row store (OLTP):

```
[id:1,name:A,amt:50] [id:2,name:B,amt:30] [id:3,name:C,amt:70] …
```

Columnar (OLAP):

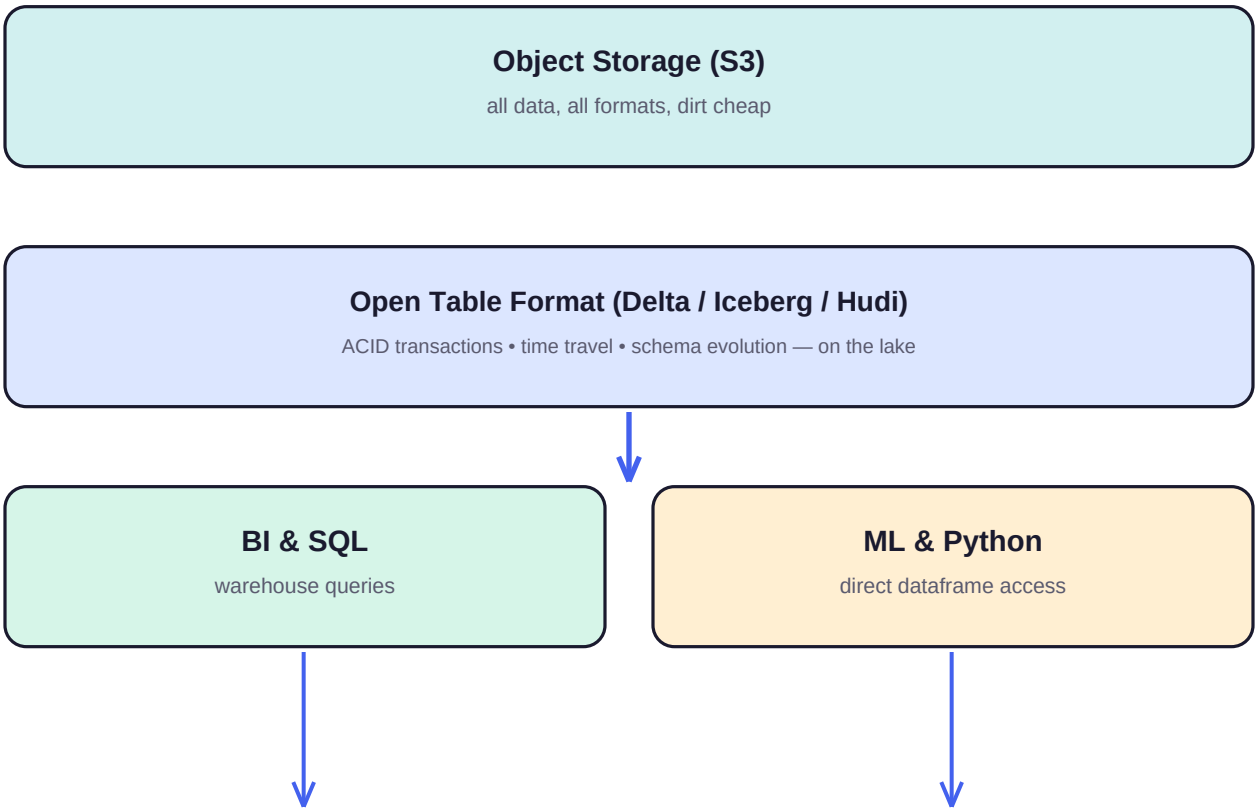
```
ids:   [1,2,3,…]   names: [A,B,C,…]   amts: [50,30,70,…]
```

- Query 'SUM(amt)': read ONE column, skip the rest — 10-100× less I/O.
- Same-type values compress beautifully (run-length, dictionary encoding).
- Parquet/ORC: the open columnar file formats. Partition by date for pruning.

STEP 4/6

Lakehouse: Lake + Warehouse

Cheap object storage with warehouse superpowers.



One copy of data serving both SQL analysts and ML engineers.

STEP 5/6

Time Travel & Schema Evolution

Superpowers tables didn't used to have.

Time travel

Query data AS OF last Tuesday. Debug, audit, recover from bad writes.

Schema evolution

Add/rename columns without rewriting petabytes.

Partition evolution

Change partition layout without breaking old queries (Iceberg).

ACID

Concurrent writers don't corrupt readers. Snapshots isolate.

STEP 6/6

Choosing Your Platform

Match the tool to the job.

Cloud warehouses

Snowflake, BigQuery, Redshift — serverless SQL, separate storage/compute.

Lakehouse

Databricks, open Iceberg/Delta on your own storage — engine freedom.

Self-managed

ClickHouse, DuckDB, Trino — control + cost at scale.

Start simple (one warehouse). Add lakehouse when ML + BI must share data at scale.

STEP ✓I✓

Key Takeaways

Warehouses & Lakehouses — Remember

- OLTP = transactions (rows); OLAP = analytics (columns). Never mix the workloads.
- Star schemas: central facts, surrounding dimensions — denormalize for speed.
- Columnar storage (Parquet) reads only needed columns — massive I/O savings.
- Lakehouse = warehouse features (ACID, time travel) on cheap object storage.
- One data copy can serve both SQL analysts and ML pipelines.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev